

Logarithmic Terminal-Window Accuracy Thresholds for the Complete Upper Counterloop Band

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Abstract

We invert the deterministic terminal-tail law of RH-358 at polynomial target accuracy. Let $E_k(q) = P_k(q)/C_k$ be the relative strict-upper-band mass left after retaining the top q terminal coordinates, and let $x > 1$ be the source-locked radial constant. For $a > 0$, $c \in \mathbb{R}$, put

$$t_k = \frac{a \log k}{\log x} + c, \quad q_k = \lfloor t_k \rfloor, \quad \theta_k = \{t_k\}.$$

We prove

$$k^a E_k(q_k) = x^{\theta_k - c} \{1 + o(1)\}.$$

The complete limit set of θ_k is $[0, 1]$, so the normalized error has the complete cluster interval $[x^{-c}, x^{1-c}]$ rather than a unique constant. If

$$Q_k(a, c) = \min\{q : E_k(q) \leq x^{-c} k^{-a}\},$$

then $Q_k = t_k + O(1)$ and the complete limit set of $Q_k - t_k$ is exactly $[0, 1]$. We also classify arbitrary sublinear windows by their polynomial accuracy exponent and translate logarithmic width to the physical double-log clock. Actual-head inheritance is only conditional on the original unnormalized same-clock leaf $D_{4k}(R) \rightarrow 0$. No root or rank identification, determinant closure, Gate A–E promotion, Hilbert–Polya construction, zero identification, or proof of RH is claimed.

1 Source-locked tail ledger

Let $k \rightarrow \infty$ be the RH-342 first-alias clock and let

$$x = (\beta R)^2 > 1, \quad y_k = (\beta_k R)^2 = x \exp \left[-\frac{\log C_M}{k} + o(k^{-1}) \right], \quad C_M > 0. \quad (1)$$

The deterministic strict upper-counterloop budgets of RH-357–RH-358 are

$$B_k(L) = \sum_{j=1}^L \frac{y_k^{k+j}}{k+j}, \quad 1 \leq L \leq k-1, \quad (2)$$

$$C_k = B_k(k-1), \quad P_k(q) = B_k(k-1-q), \quad 0 \leq q \leq k-2. \quad (3)$$

Define the relative truncation error

$$E_k(q) := \frac{P_k(q)}{C_k}. \quad (4)$$

Equivalently, $1 - E_k(q)$ is the normalized mass in the top q terminal coordinates. Since every terminal-lag weight is positive, $E_k(q)$ is strictly decreasing in q .

RH-358 proves the uniform source-locked formula

$$E_k(q) = x^{-q} C_M^{q/k} \frac{2k-1}{2k-1-q} \frac{1-x^{-(k-1-q)}}{1-x^{-(k-1)}} \{1 + o(1)\}, \quad (5)$$

where the relative $o(1)$ is uniform over $0 \leq q \leq k - 2$ [4]. The finite-tail factor in (5) is essential near $q = k - 2$. The present paper uses only the logarithmic subrange, where every displayed correction can be quantified without deleting it outside its valid domain.

The symbol q here is an integer window width. It is not the open direct/full-trace quantity denoted by the same letter elsewhere in the project.

2 Uniform logarithmic reduction and phase density

Lemma 2.1 (Uniform logarithmic window). *For every fixed $A > 0$,*

$$\sup_{0 \leq q \leq A \log k} |x^q E_k(q) - 1| \longrightarrow 0, \quad (6)$$

where the supremum is over admissible integers.

Proof. In the stated range, $q/k = O(\log k/k)$. Hence $C_M^{q/k} \rightarrow 1$ and $(2k - 1)/(2k - 1 - q) \rightarrow 1$ uniformly. Also $k - 1 - q \rightarrow \infty$ uniformly, so both finite-tail factors in (5) tend uniformly to one. The source remainder in (5) is already uniform on the full admissible range. Multiplication by x^q proves (6). $\square$

The logarithmic phase mechanism was used for a different post-alias ratio in RH-356 [3]. We record the exact generality needed for the accuracy problem.

Lemma 2.2 (Complete phase set). *Fix $a > 0$ and $c \in \mathbb{R}$. Then*

$$\theta_k(a, c) := \left\{ \frac{a \log k}{\log x} + c \right\} \quad (7)$$

has complete limit set $[0, 1]$.

Proof. For $t \in (0, 1)$, take

$$k_m = \left\lfloor x^{(m+t-c)/a} \right\rfloor.$$

Then $a \log_x k_m + c = m + t + o(1)$, so a subsequence of phases tends to t . For the endpoints, use

$$k_m^{(0)} = \left\lceil x^{(m-c)/a} \right\rceil, \quad k_m^{(1)} = \left\lceil x^{(m+1-c)/a} \right\rceil - 1.$$

The first phase tends to zero and the second to one. Subtracting one in the second construction handles exact integer powers. No phase outside $[0, 1]$ is possible. $\square$

3 Polynomial accuracy and its integer phase

Fix $a > 0$ and $c \in \mathbb{R}$, and define

$$t_k(a, c) := \frac{a \log k}{\log x} + c, \quad q_k(a, c) := \lfloor t_k(a, c) \rfloor, \quad \theta_k(a, c) := \{t_k(a, c)\}. \quad (8)$$

These integers are admissible for all sufficiently large k .

Theorem 3.1 (Polynomial phase law). *With (8),*

$$\boxed{k^a E_k(q_k(a, c)) = x^{\theta_k(a, c) - c} \{1 + o(1)\}}. \quad (9)$$

The complete limit set of the left-hand side is

$$\boxed{[x^{-c}, x^{1-c}]}. \quad (10)$$

In particular,

$$\liminf k^a E_k(q_k) = x^{-c}, \quad \limsup k^a E_k(q_k) = x^{1-c}, \quad (11)$$

and the sequence has no single limit.

Proof. Since $q_k = O(\log k)$, lemma 2.1 gives $E_k(q_k) = x^{-q_k}\{1 + o(1)\}$. The identity

$$q_k = \frac{a \log k}{\log x} + c - \theta_k$$

then yields (9). Apply lemma 2.2 and the continuous increasing map $t \mapsto x^{t-c}$ to obtain the complete cluster interval. Its endpoints differ because $x > 1$. $\square$

Corollary 3.2 (Accuracy-exponent classification). *Let $q_k = o(k)$ be admissible. Then*

$$\log E_k(q_k) = -q_k \log x + o(1). \quad (12)$$

Consequently, if

$$\frac{q_k \log x}{\log k} \longrightarrow a \in [0, \infty], \quad (13)$$

then

$$\frac{\log E_k(q_k)}{\log k} \longrightarrow -a. \quad (14)$$

For $a = \infty$, this means $E_k(q_k) = o(k^{-A})$ for every fixed $A > 0$.

Proof. The sublinear specialization of (5) is $E_k(q_k) = x^{-q_k}\{1 + o(1)\}$. Taking logarithms proves (12); division by $\log k$ gives the finite cases. If the ratio in (13) diverges, then for every fixed A it eventually exceeds $A + 1$, and (12) gives the asserted superpolynomial bound. $\square$

4 The exact minimal integer window

For $a > 0$ and $c \in \mathbb{R}$, let the target error be

$$\varepsilon_k(a, c) := x^{-c} k^{-a} = x^{-t_k(a, c)}. \quad (15)$$

Because $E_k(0) = 1$, E_k is strictly decreasing, and the final admissible tail is exponentially small, the following minimum exists for all large k :

$$Q_k(a, c) := \min\{0 \leq q \leq k - 2 : E_k(q) \leq \varepsilon_k(a, c)\}. \quad (16)$$

Theorem 4.1 (Sharp inverse accuracy theorem). *For every $a > 0$ and $c \in \mathbb{R}$,*

$$Q_k(a, c) = t_k(a, c) + O(1). \quad (17)$$

More precisely, the complete limit set of

$$Q_k(a, c) - t_k(a, c) \quad (18)$$

is exactly

$$\boxed{[0, 1]}. \quad (19)$$

If $\theta_k(a, c)$ remains in a fixed compact subinterval of $(0, 1)$ along a subsequence, then on that subsequence

$$Q_k(a, c) = \lceil t_k(a, c) \rceil \quad (20)$$

eventually.

Proof. Write $n_k = \lfloor t_k \rfloor$ and $\theta_k = \{t_k\}$. For each fixed $j \in \{-1, 0, 1, 2\}$, lemma 2.1 gives

$$\frac{E_k(n_k + j)}{\varepsilon_k(a, c)} = x^{\theta_k - j} \{1 + o(1)\}, \quad (21)$$

whenever the index is admissible, which it is eventually. The case $j = -1$ is eventually greater than one, while $j = 2$ is eventually less than one. Thus

$$n_k \leq Q_k \leq n_k + 2, \quad (22)$$

proving (17).

Consider any convergent subsequence of (18). Passing to a further subsequence, $Q_k - n_k$ is one fixed value $j \in \{0, 1, 2\}$ and $\theta_k \rightarrow \theta \in [0, 1]$. If $j = 0$, minimality and (21) force $x^\theta \leq 1$, hence $\theta = 0$ and the correction tends to zero. If $j = 2$, then $E_k(n_k + 1) > \varepsilon_k$; (21) forces $x^{\theta-1} \geq 1$, hence $\theta = 1$ and the correction tends to one. If $j = 1$, the correction tends to $1 - \theta \in [0, 1]$. Therefore every cluster point lies in $[0, 1]$.

Conversely, fix $s \in (0, 1)$. By lemma 2.2, choose a subsequence with $\theta_k \rightarrow 1 - s$. Along it, the $j = 0$ ratio in (21) tends to $x^{1-s} > 1$, while the $j = 1$ ratio tends to $x^{-s} < 1$. Hence $Q_k = n_k + 1$ eventually and $Q_k - t_k \rightarrow s$. The limit set of a bounded sequence is closed, so it also contains the endpoints. This proves (19). The same strict two-ratio comparison proves (20) whenever the phase is bounded away from zero and one. $\square$

Remark 4.2 (Why no stronger universal integer formula is claimed). At a phase endpoint, one of the ratios in (21) tends to one. The source lock (1) allows an unspecified $o(k^{-1})$ term, so its sign may decide between adjacent integers. The interval (19) is the sharp source-backed conclusion.

5 Physical clock and conditional actual-head inheritance

On the RH-355 physical clock,

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1). \quad (23)$$

Therefore

$$\frac{a \log k}{\log x} + c = \frac{a}{\log x} \log \log(1/\sigma) + c - \frac{a}{\log x} \log(2 \log \lambda) + o(1). \quad (24)$$

Polynomial-in- k truncation accuracy thus requires a double-logarithmic number of terminal coordinates. Equation (19) prevents promotion of (24) to a unique integer constant without a phase subsequence.

Let $E_k^{\mathcal{H}}(q)$ be the corresponding actual even-weight relative tail from RH-358. Under the original same-clock unnormalized hypothesis

$$D_{4k}(R) := \sum_{2 \leq n < 4k} \frac{|h_{\sigma,n} - s_{k,n}| R^n}{n} \longrightarrow 0, \quad (25)$$

RH-358 proves

$$\sup_{0 \leq q \leq k-2} \left| \frac{E_k^{\mathcal{H}}(q)}{E_k(q)} - 1 \right| \longrightarrow 0. \quad (26)$$

Theorem 5.1 (Conditional actual accuracy inheritance). *Assume (25) on one physical clock, and define*

$$Q_k^{\mathcal{H}}(a, c) := \min\{0 \leq q \leq k - 2 : E_k^{\mathcal{H}}(q) \leq x^{-c} k^{-a}\}.$$

Then the phase law (9) and exponent law (14) hold with $E_k^{\mathcal{H}}$ in place of E_k , and the complete limit set of $Q_k^{\mathcal{H}}(a, c) - t_k(a, c)$ is $[0, 1]$.

Proof. Equation (26) multiplies every tail ratio used above by $1 + o(1)$ uniformly in q . The proofs of theorems 3.1 and 4.1 and corollary 3.2 use only such ratios, monotonicity of nonnegative tails, and phase subsequences. Repeating them gives the conditional claims. $\square$

This theorem is inactive until (25) is proved. It transfers weighted budget accuracy only, not roots, ranks, eigenvalues, or determinants.

6 Finite protocol, boundary, and next route

The artifact uses the exact rational fixture $x = 2352/1445$ to evaluate exact terminal tails and exact minimal widths for rational polynomial targets. High-precision rows reproduce the phase formula and finite cluster coverage. They are finite checks, not evidence for phase density or an actual noisy operator.

The next read-only candidate is RH-360: the exponential generating function $G_k(z) = \sum_r z^r \pi_k(r)$ across the critical value $z = x$. The anticipated scales are a finite transform below x , a k -scaled Riemann-integral law in the critical window $z_k = x \exp(\tau/k)$, and exponential endpoint dominance above x . No such candidate is pre-authorized as an actual spectral law.

RH-359 does not prove (25); identify actual roots, ranks, or spectra; close a determinant, p , the unrelated open q , or E_{off} ; close the moving noisy envelope of RH-241 [1]; or activate the gluing criterion of RH-288 [2]. Gates A–E remain false/open. There is no Hilbert–Polya operator, Riemann-zero identification, von Mangoldt trace theorem, completed-zeta divisor equality, or proof of RH.

References

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